

Aayush Bisht

✉ aayush.bisht@student-cs.fr | 📞 +33 7 45 39 97 29 | 🔗 linkedin.com/in/aayush-bisht | 🐙 github.com/aayushbisht

AI/ML Engineer & Data Scientist

EXPERIENCE

Amadeus

AI Engineer Intern

Sophia Antipolis, France

May 2026 – Oct 2026

- Built an Agentic AI documentation generator using multi-agent orchestration to produce documentation from any repository, drastically reducing manual effort (15 min per repo) and cost (\$8 to \$2), with another multi-agent system verifying output for accuracy and reliability.
- Developed an AI chatbot grounded in the generated documentation, enabling instant retrieval of queries across teams.

Cybercell

Cybersecurity Analyst (OSINT)

Kota, India

Aug 2023 – Oct 2024

- Engineered automated monitoring scripts (Python) to filter high-volume threat data streams, reducing manual triage time by approximately 30%.
- Conducted multimodal analysis using OSINT frameworks to trace digital footprints and identify security breaches in real-time.

QuantStreet LLP

Data Analyst

New Delhi, India

Aug 2019 – Jul 2023

- Engineered automated SQL and Pandas data pipelines, eliminating 12+ hours of manual processing per week and mitigating noise in high-frequency trading datasets to drive a 4% accuracy improvement in strategy optimization models.

EDUCATION

CentraleSupélec, University Paris-Saclay

MSc in Artificial Intelligence

Paris, France

Sep 2025 – Present

- Relevant Coursework:** Machine Learning, Deep Learning, Optimization, Natural Language Processing, Decision Modeling, AI for Finance, Medical Imaging (Computer Vision), Multi-Agent System.
- MVA, ENS (École Normale Supérieure):** Reinforcement Learning, Graphs in Machine Learning, Advanced Deep Learning.

Graphic Era University

Bachelor of Technology, Information Technology

Dehradun, India

2015 – 2019

KEY AI PROJECTS

Causal Discovery in Time Series for Trustworthy AI

CEA Lab (Research Project)

- Benchmarked causal discovery algorithms (PCMCI, PCMCI+) against the Causal Chamber dataset to evaluate ground-truth recovery.
- Stress-tested model robustness using **Tigramite**, analyzing F1-scores and Structural Hamming Distance (SHD) under conditions of hidden confounding and autocorrelation.

Hierarchical Ensemble for Text Decoding

CentraleSupélec

- Developed a hybrid architecture combining a **GRU-Seq2Seq** model for decryption and a fine-tuned **GPT-2** for linguistic refinement.
- Implemented an end-to-end inference pipeline in **PyTorch**, successfully reconstructing encrypted text with high semantic accuracy.

Decoding mouse position in maze from brain signals.

Paris Brain Institute

- Developed deep-learning models to decode a mouse's spatial position from hippocampal spike waveforms, evaluating input representations, temporal windows, and data-splitting strategies for robust generalization.

RL-Based Neural Network Pruning for Causal Time Series Models

CentraleSupélec

- Engineered a **Deep Q-Network (DQN)** agent to automate layer-wise pruning of **Temporal Convolutional Networks**, achieving a 38.7% parameter reduction (1.63x compression) while preserving predictive accuracy and causal robustness.

Skin Cell Culture Monitoring System

2nd place Hack'In Saclay, IP Paris

- Built a **real-time monitoring dashboard** for skin-cultivation labs that scores culture health from simulated temperature, pH, O2, and impedance data, and pairs it with a **RAG chatbot** over lab SOPs plus automated email alerts carrying **AI-generated remediation steps**.

TECHNICAL SKILLS

Languages: Python, SQL.

AI/ML Frameworks: PyTorch, TensorFlow, Azure OpenAI, HuggingFace, LLM APIs, Multi-Agent System, LangChain, LangGraph, RAG (Retrieval-Augmented Generation), Ollama, Strands Agents, Pandas, NumPy.

Data & Tools: DVC, Git, Docker, JIRA, Splunk, Linux, ETL, Confluence.

Languages: English, French (A2), Hindi.